

Chapter 9: Project Human Resource Management

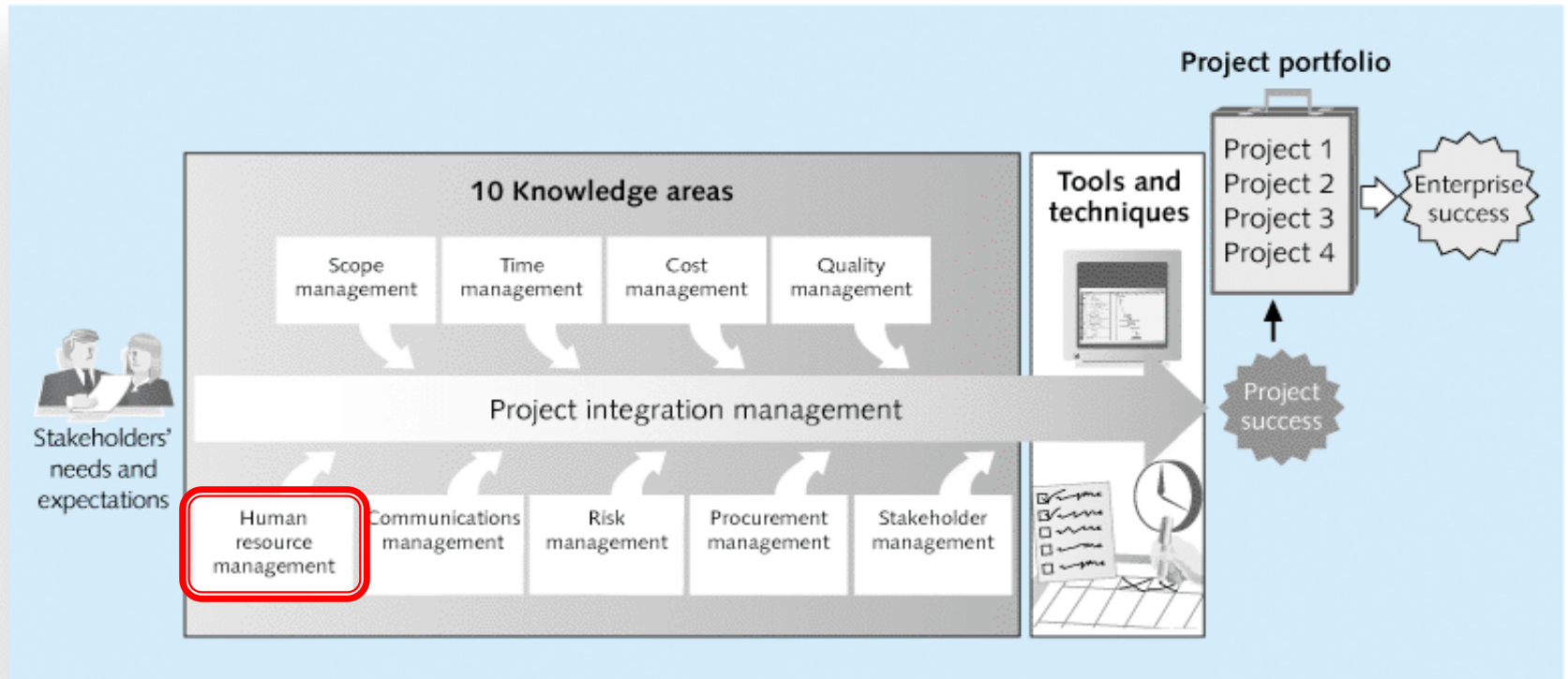
Information Technology Project
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Information Technology
PROJECT MANAGEMENT | 7e

Kathy Schwalbe

ĐẶNG THỊ THU HÀ – SE
dtthuha79@gmail.com
<https://lms.iuh.edu.vn>



Learning Objectives

- ▶ Explain the importance of good human resource management on projects
- ▶ Define project human resource management and understand its processes
- ▶ Understand important issues involved in project staff acquisition and explain the concepts of resource assignments, resource loading, and resource leveling
- ▶ Assist in team development with training, team-building activities, and reward systems

The Importance of Human Resource Management

- ▶ Many corporate executives have said, “People are our most important asset”
- ▶ People determine the success and failure of organizations and projects

What is Project Human Resource Management?

9.1.Planning human resource management: identifying and documenting project roles, responsibilities, and reporting relationships

9.2.Acquiring the project team: getting the needed personnel assigned to and working on the project

9.3.Developing the project team: building individual and group skills to enhance project performance

9.4.Managing the project team: tracking team member performance, motivating team members, providing timely feedback, resolving issues and conflicts, and coordinating changes to help enhance project performance.

Figure 9-1. Project Human Resource Management Summary

Planning

Process: **Plan human resource management**

Output: Human resource plan

Executing

Process: **Acquire project team**

Outputs: Project staff assignments, resource calendars, project management plan updates

Process: **Develop project team**

Outputs: Team performance assessments, enterprise environmental factors updates

Monitoring and Controlling

Process: **Manage project team**

Outputs: Change requests, project management plan updates, project documents updates, enterprise environmental factors updates, and organizational process assets updates

Project Start

Project Finish

Keys to Managing People

- ▶ Important areas related to project management include
 - motivation theories
 - influence and power
 - effectiveness

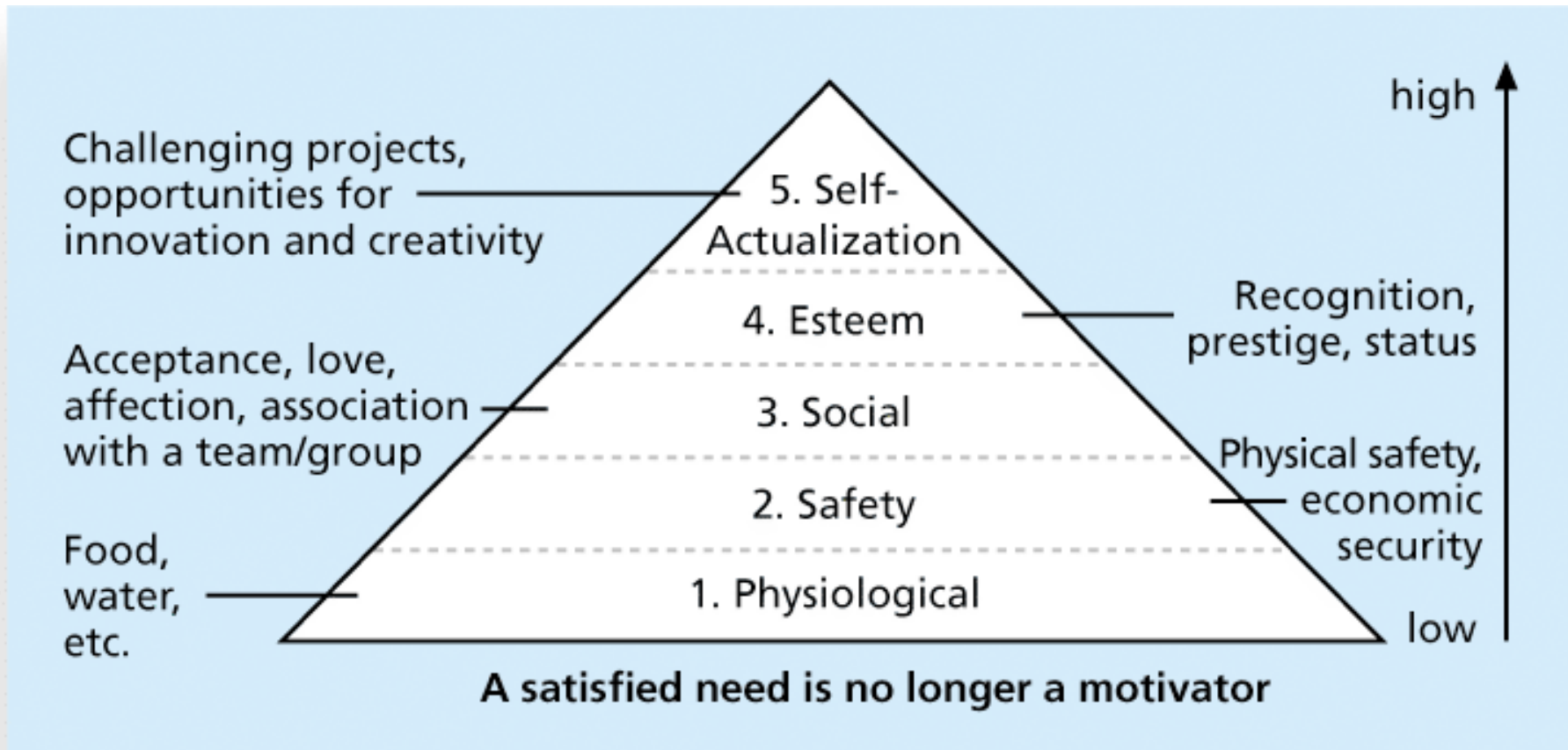
Intrinsic and Extrinsic Motivation

- ▶ **Intrinsic motivation** causes people to participate in an activity for their own enjoyment
- ▶ **Extrinsic motivation** causes people to do something for a reward or to avoid a penalty
- ▶ For example, some children take piano lessons for intrinsic motivation (they enjoy it) while others take them for extrinsic motivation (to get a reward or avoid punishment)

Maslow's Hierarchy of Needs

- ▶ Abraham Maslow argued that humans possess unique qualities that enable them to make independent choices, thus giving them control of their destiny
- ▶ Maslow developed a **hierarchy of needs** which states that people's behaviors are guided or motivated by a sequence of needs

Figure 9-2. Maslow's Hierarchy of Needs



Herzberg's Motivational and Hygiene Factors

- ▶ Frederick Herzberg wrote several famous books and articles about worker motivation. He distinguished between
 - motivational factors: achievement, recognition, the work itself, responsibility, advancement, and growth, which produce job satisfaction
 - hygiene factors: cause dissatisfaction if not present, but do not motivate workers to do more.

Examples include larger salaries, more supervision, and a more attractive work environment

9.1. Developing the Human Resource Plan

- ▶ Involves identifying and documenting project roles, responsibilities, and reporting relationships
- ▶ Contents include
 - project organizational charts
 - staffing management plan
 - responsibility assignment matrixes
 - resource histograms

Figure 9-3. Sample Organizational Chart for a Large IT Project

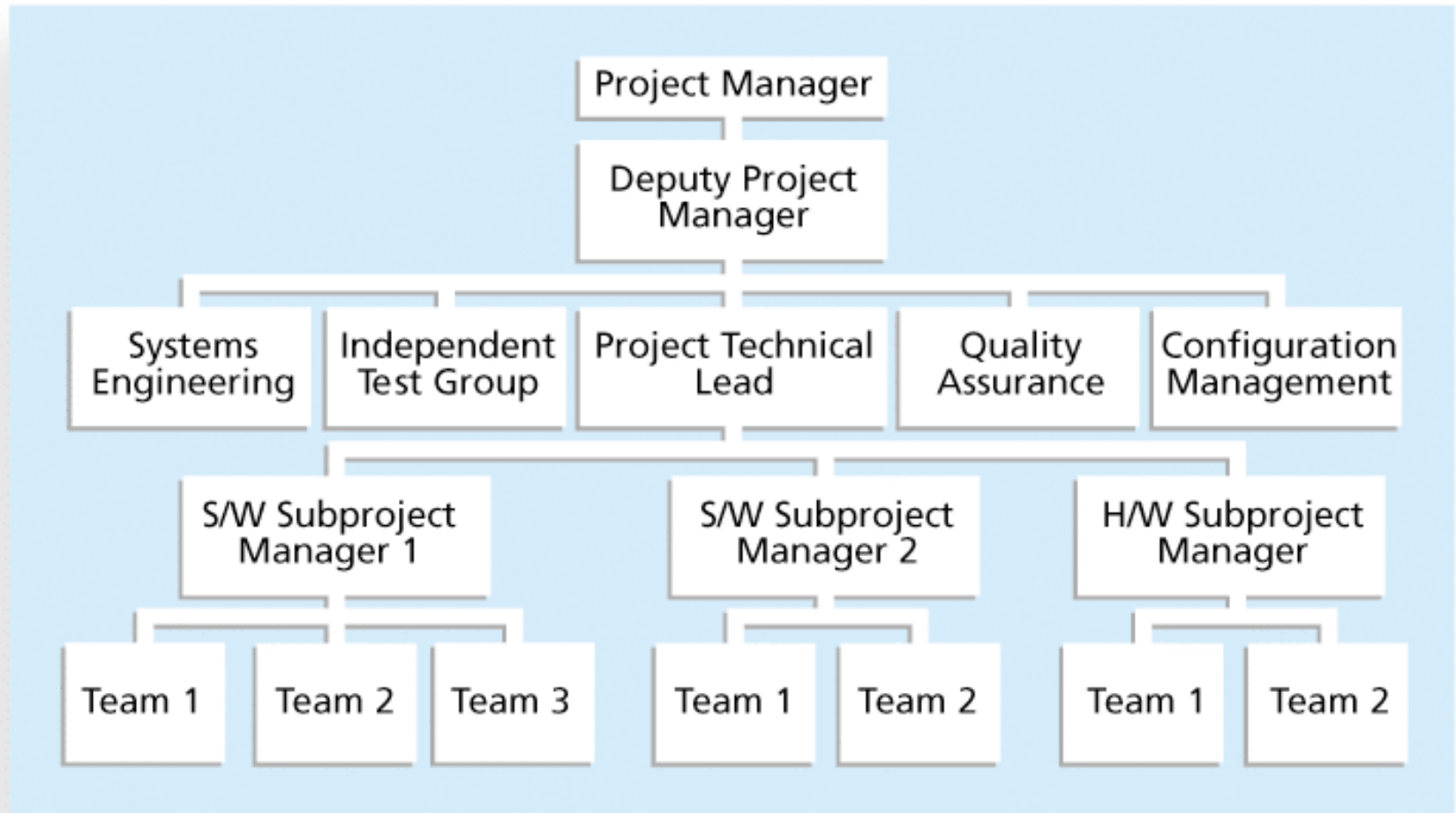
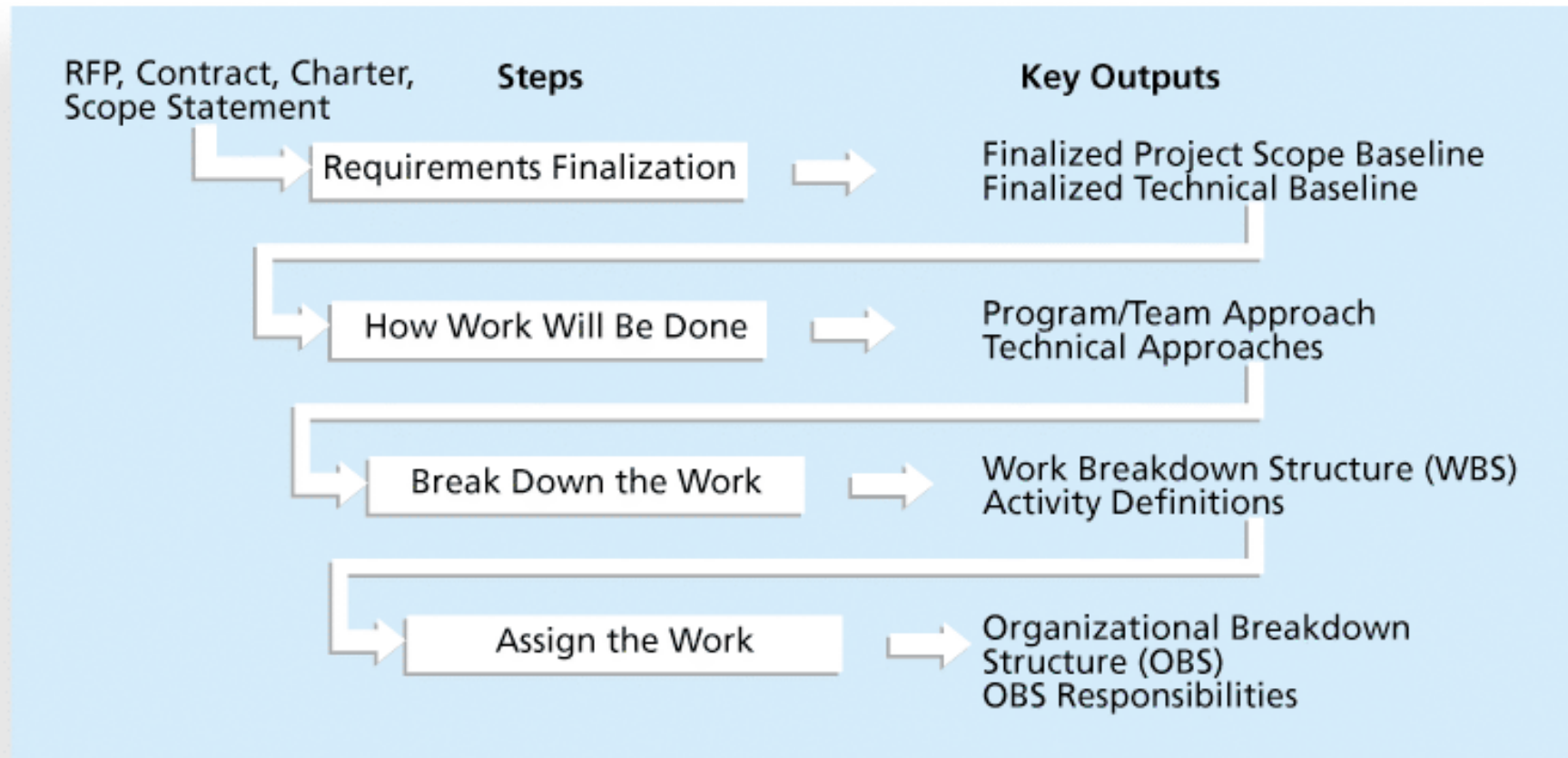


Figure 9-4. Work Definition and Assignment Process



Responsibility Assignment Matrixes

- ▶ A **responsibility assignment matrix (RAM)** is a matrix that maps the work of the project as described in the WBS to the people responsible for performing the work as described in the OBS
- ▶ Can be created in different ways to meet unique project needs

Figure 9-5. Sample Responsibility Assignment Matrix (RAM)

WBS activities →

OBS units ↓

	1.1.1	1.1.2	1.1.3	1.1.4	1.1.5	1.1.6	1.1.7	1.1.8
Systems Engineering	R	R P					R	
Software Development			R P					
Hardware Development				R P				
Test Engineering	P							
Quality Assurance					R P			
Configuration Management						R P		
Integrated Logistics Support							P	
Training								R P

R = Responsible organizational unit

P = Performing organizational unit

RACI charts - Detailed RAM

- ▶ **RACI charts** to show four key roles for project stakeholders:

Responsibility: Who does the task?

Accountability: Who signs off on the task or has authority for it?

Consultation: Who has information necessary to complete the task?

Informed: Who needs to be notified of task status and results?

Table 9-2. Sample RACI Chart

	Group A	Group B	Group C	Group D	Group E
Test Plans	R	A	C	C	I
Unit Test	C	I	R	A	I
Integration Test	A	R	I	C	C
System Test	I	C	A	I	R
User Acceptance Test	R	I	C	R	A

R = responsibility

A = accountability, only one A per task

C = consultation

I = informed

Note that some people reverse the definitions of responsible and accountable.

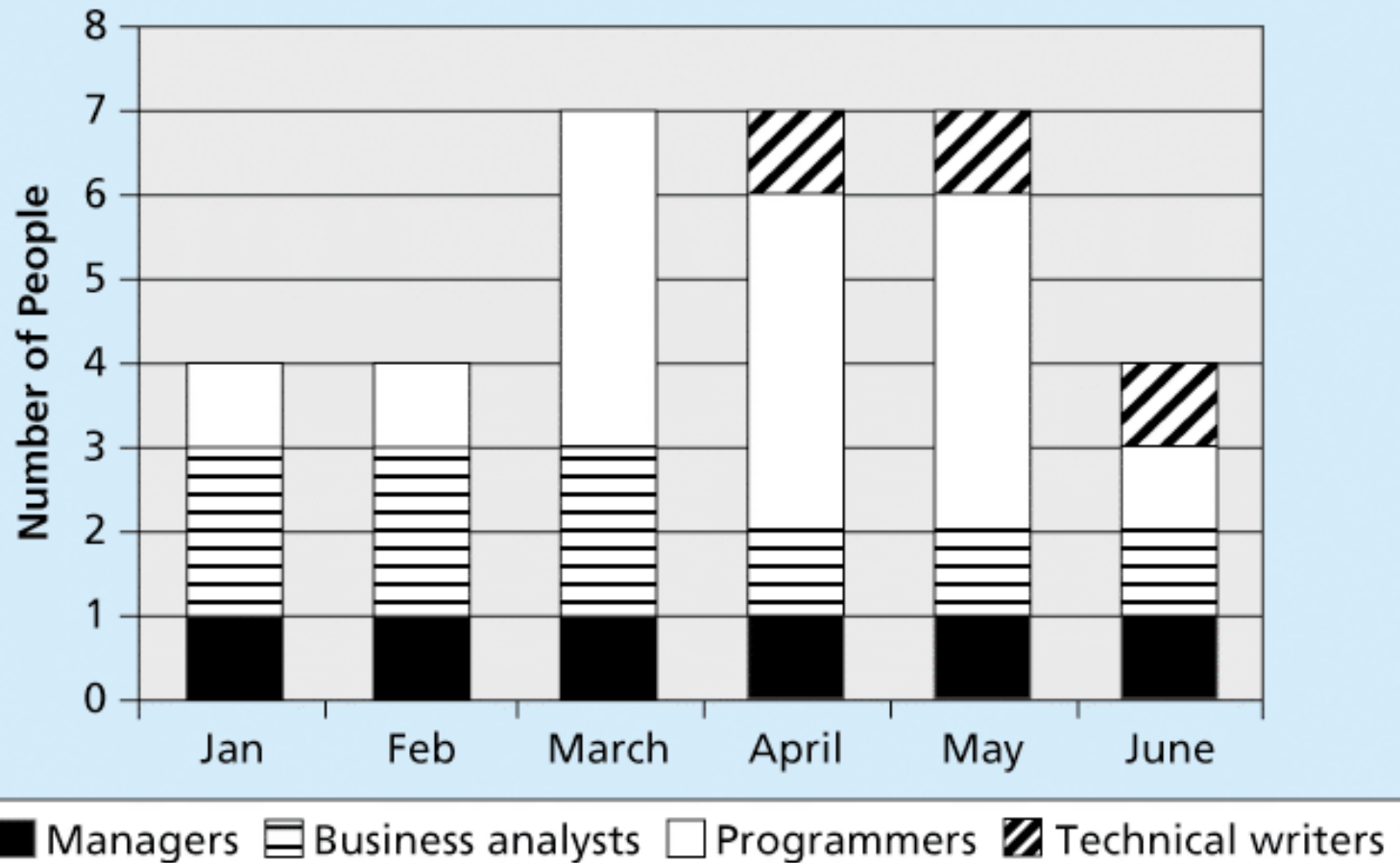
Function	Project Sponsor	Project Manager	Developer	Analyst
Project Initiation	R,P	R,P		
Establish Project Plan		R,P		
Collate User Requirements				R,P
Define Technical Requirements				R,P
Develop Software Tools			R,P	
Test Software			R,P	
Install Software			R,P	

Function	Project Sponsor	Project Manager	Developer	Analyst
Project Initiation	C	A,R		
Establish Project Plan	I	A,R	C	C
Collate User Requirements	I	A	I	R
Define Technical Requirements	I	A		R
Develop Software Tools	I	A	R	C
Test Software	I	A	C	R
Install Software	C	A	C	R

Staffing Management Plans and Resource Histograms

- ▶ A **staffing management plan** describes when and how people will be added to and taken off the project team
- ▶ A **resource histogram** is a column chart that shows the number of resources assigned to a project over time

Figure 9-6. Sample Resource Histogram



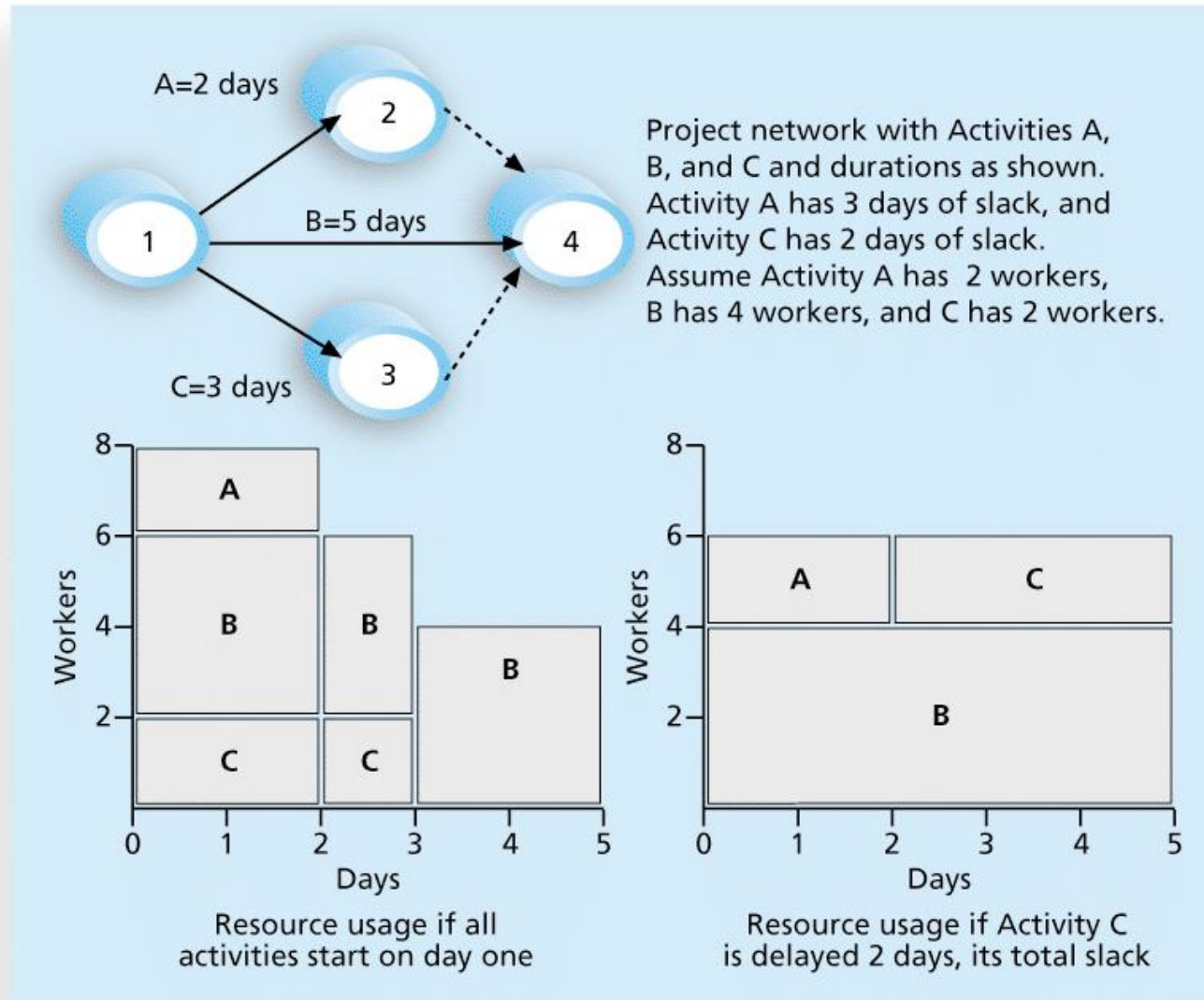
9.2.Acquiring the Project Team

- ▶ Acquiring qualified people for teams is crucial
- ▶ The project manager who is the smartest person on the team has done a poor job of recruiting!
- ▶ It's important to assign the appropriate type and number of people to work on projects at the appropriate times

Resource Leveling

- ▶ **Resource leveling** is a technique for resolving resource conflicts by delaying tasks
- ▶ The main purpose of resource leveling is to create a smoother distribution of resource usage and reduce overallocation

Figure 9-8. Resource Leveling Example



Benefits of Resource Leveling

- ▶ When resources are used on a more constant basis, they require less management
- ▶ It may enable project managers to use a just-in-time inventory type of policy for using subcontractors or other expensive resources
- ▶ It results in fewer problems for project personnel and accounting department
- ▶ It often improves morale

9.3.Developing the Project Team

- ▶ The main goal of **team development** is to help people work together more effectively to improve project performance
- ▶ It takes teamwork to successfully complete most projects

Model of Team Development

- ▶ Forming
- ▶ Storming
- ▶ Norming
- ▶ Performing
- ▶ Adjourning

Training

- ▶ Training can help people understand themselves, each other, and how to work better in teams
- ▶ Team building activities include
 - physical challenges
 - psychological preference indicator tools

9.4.Managing the Project Team

- ▶ Project managers must lead their teams in performing various project activities
- ▶ After assessing team performance and related information, the project manager must decide
 - if changes should be requested to the project
 - if corrective or preventive actions should be recommended
 - if updates are needed to the project management plan or organizational process assets.

Tools and Techniques for Managing Project Teams

- ▶ Observation and conversation
- ▶ Project performance appraisals
- ▶ Interpersonal skills
- ▶ Conflict management

Conflict Handling Modes

1. **Confrontation:** Directly face a conflict using a problem-solving approach
2. **Compromise:** Use a give-and-take approach
3. **Smoothing:** De-emphasize areas of difference and emphasize areas of agreement
4. **Forcing:** The win-lose approach
5. **Withdrawal:** Retreat or withdraw from an actual or potential disagreement
6. **Collaborating:** Decision makers incorporate different viewpoints and insights to develop consensus and commitment

Conflict Can Be Good

- ▶ Conflict often produces important results, such as new ideas, better alternatives, and motivation to work harder and more collaboratively
- ▶ **Groupthink:** Conformance to the values or ethical standards of a group. Groupthink can develop if there are no conflicting viewpoints
- ▶ Research suggests that task-related conflict often improves team performance, but emotional conflict often depresses team performance

Five Dysfunctions of a Team

- ▶ The five dysfunctions of teams are
 1. Absence of trust
 2. Fear of conflict
 3. Lack of commitment
 4. Avoidance of accountability
 5. Inattention to results

*Lencioni, Patrick, “Overcoming the Five Dysfunctions of a Team,” Jossey-Bass: San Francisco, CA (2005), p. 3.

Chapter Summary

- ▶ Project human resource management includes the processes required to make the most effective use of the people involved with a project
- ▶ Main processes include
 - Plan human resource management
 - Acquire project team
 - Develop project team
 - Manage project team